

The physical properties of two isoforms of a protein, both of which bind the same ligand, are shown in the table:

Isoform	Molecular weight (kDa)	pI	K_d (M)
1	68.3	4.7	3.1×10^{-6}
2	67.7	5.9	2.9×10^{-6}

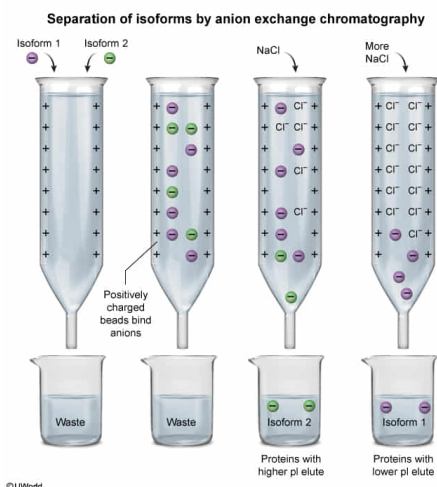
Which protocol would yield the best separation of the two isoforms?

- ☐ A. Run the mixture through a cation-exchange column starting at pH 10, then gradually increase the pH of the buffer. [23%]
- ☐ B. Pass the mixture through a size-exclusion column with a pore size cutoff of 40 kDa. [1%]
- ☒ C. Load the mixture onto an anion-exchange column at physiological pH, then gradually increase the NaCl concentration of the buffer. [59%]
- ☐ D. Load the mixture onto an affinity column with the ligand covalently bound to the beads, followed by cleaving the covalent bond to elute. [15%]

Correct

59% answered correctly

Explanation:



Separation techniques take advantage of molecular properties to cause some molecules to move faster than others through a **stationary phase**. As the molecules exit the stationary phase (elute), they can be collected at different time points. Column chromatography is a set of techniques for separating proteins by **charge (ion exchange)**, **molecular weight (size exclusion)**, or **binding partner (affinity)**.

The charge of a protein depends on the pH at which the protein has a **net charge of zero**, which is known as the **isoelectric point** (pI). When the ambient pH is lower than the pI, the protein gains protons and becomes positively charged. When pH is greater than the pI, the protein loses protons and becomes negatively charged.

Both isoforms shown in the table have pI values below physiological pH (7.4), so at this pH, both are **negatively charged** and will bind to an **anion-exchange** column. When NaCl is added, the **chloride anions compete** with the negatively charged proteins for binding. Because Isoform 1 has a lower pI than Isoform 2, it is **more negatively charged** and **binds the column more tightly**. Therefore, Isoform 2 will elute at a lower salt concentration than Isoform 1. Gradually increasing the salt concentration in the buffer allows each isoform to be collected separately.

(Choice A) **Cation-exchange** columns have negatively charged beads that bind positively charged molecules and repel negatively charged ones. At pH 10, both isoforms are negatively charged and would be repelled by the beads, exiting the column together.

(Choice B) Both isoforms are approximately the same size (~68 kDa), so they will elute from the **size-exclusion** column together.

(Choice D) Both isoforms bind the same ligand with nearly identical affinity ($\sim 3 \times 10^{-6}$ M), so both would bind to the [affinity column](#). This technique would separate the two isoforms from other proteins but not from each other.

Educational objective:

Proteins are often purified on chromatography columns. Ion-exchange columns separate molecules by charge (cation-exchange columns bind cations; anion-exchange columns bind anions), size-exclusion columns separate molecules by size, and affinity columns isolate molecules that bind a particular ligand.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402300

System:
Amino Acids and Proteins